

Takeoff Data Card - Turnback Analysis

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C150 Flatwing - KSEZ - RWY 21

Weight 1,500 lb - Field 4,827 ft MSL - ISA10°C - Wind 5 kt from 210°

475 ft AGL

Recommended turnback minimum (x 1.25 safety factor) - LEFT turn - 5,302 ft MSL on the altimeter

Aircraft & Conditions

Aircraft	C150 Flatwing
Gross weight	1,500 lb (MTOW 1,600 lb)
Airport	KSEZ
Runway	21
Field elevation	4,827 ft MSL
ISA deviation	10 °C
Altimeter	29.92 inHg
Surface wind	5 kt from 210° true (rel runway 210°)
Wind components	4.0 kt HEADwind - 2.0 kt RIGHT cross
Climb-out KIAS	68 KIAS
Takeoff flap	Clean
Flap-retract altitude	<i>n/a (clean takeoff)</i>
Turn flap	Clean
Bank angle (turnback)	45°
Turnback speed mode	Fixed 68 KIAS
Reaction time	5.0 sec

Winds Aloft

No winds-aloft profile entered.

Runway

Available runway length	5,132 ft
Liftoff distance (POH-scaled)	800 ft
Last-abort point on runway	- <i>(future feature)</i>
Straight-ahead landing limit	0 - 80 ft AGL <i>(failure below this altitude -> land straight ahead, staying within the airport boundary - not just the runway asphalt)</i>

Turnback - Critical Decision Numbers

Recommended minimum altitude	Recommended turn direction
475 ft AGL	LEFT
5,302 ft MSL on your altimeter. Calc 380 ft x 1.25 safety factor. Below this altitude on takeoff: land straight ahead.	LEFT requires only 380 ft AGL vs 410 ft AGL for RIGHT.

Critical altitude - turn LEFT	380 ft AGL (5,207 ft MSL)
Critical altitude - turn RIGHT	410 ft AGL (5,237 ft MSL)
Threshold-crossing altitude (margin)	- (-) - -
Crossing the takeoff-end threshold	- (-)
Crossing the departure-end threshold	- (-) - relevant for downwind landing case
Total degrees of turn required	0°
Altitude lost per 180° of turn	-
Turn radius (at chosen bank)	-
Landing direction	-

[!] DEAD ZONE: Engine failure between **80 ft** and **380 ft AGL** = no good options. Plan to climb through this band as quickly as possible.

Training Brief - what these numbers mean

Critical altitude is the lowest AGL at which the airplane can complete the turnback and reach the runway. Below this, the turnback fails - the airplane runs out of altitude before the runway is in reach. The LEFT/RIGHT split exists because crosswind drift, slope, and obstacles make the two directions asymmetric. **Always pick the lower of the two.**

The 180° figure is half-mythology in the cockpit. In reality the turn must overshoot the runway centerline (typically 200-270°) so the airplane is aligned on final, not perpendicular. The "Sequence of Events" table above shows the actual heading change.

Safety margin x Calculated = recommended go/no-go. The calculated number is the bare aerodynamic minimum; the recommended number (475 ft AGL) is what you brief. Below the recommended altitude on takeoff, plan to land straight ahead within the airport boundary.

Reaction time is the killer. Three seconds doubles or triples the dead-zone altitude band. Brief it on every takeoff: "engine quits below 475 ft AGL -> straight ahead, no negotiation."

Bank angle trades stall margin for turn radius. At 45° bank, stall speed is 19% higher than wings-level. Steeper than 45° is rarely worth it: the radius gain is small but the stall margin shrinks fast.

Practice Drill

1. Climb to a safe altitude (3,000 ft AGL or higher).
2. Reduce to climb-out KIAS shown above (68 KIAS), simulate idle power, wait the 5.0 sec reaction time.
3. Roll into a 45° bank turn at the indicated turn-flap setting.
4. Time a 180° turn and read the altimeter delta.
5. Compare to the - figure above. If your real loss is higher, increase your personal safety factor.

Sources: Rogers (USNA, *Looking Back at the Turn-Back Maneuver*), Jett (USAFA, DTIC ADA122862), FAA AC 61-83K para A.114, FAA AFH Ch.6, EAA Sport Aviation May 2026.

This card is a planning tool - not a substitute for the aircraft POH or pilot judgment.